Selection Sort

List Size	Comparisons	Time (seconds)
1,000 (observed)	499500	0.028
2,000 (observed)	1999000	0.065
4,000 (observed)	7998000	0.179
8,000 (observed)	31996000	0.615
16,000 (observed)	127992000	2.383
32,000 (observed)	511984000	9.497
100,000 (estimated)	4999950000	103.699
500,000 (estimated)	124999750000	2592.00
1,000,000 (estimated)	499999500000	10370.00
10,000,000 (estimated)	49999995000000	1036990.00

Insertion Sort

List Size	Comparisons	Time (seconds)
1,000 (observed)	249067	0.023
2,000 (observed)	992890	0.059
4,000 (observed)	3999429	0.151
8,000 (observed)	161151116	0.513
16,000 (observed)	64321667	2.061
32,000 (observed)	25745590	8.006
100,000 (estimated)	2492405821	82.659
500,000 (estimated)	62499875000	2066.00
1,000,000 (estimated)	249999750000	8266
10,000,000 (estimated)	24999997500000	826,590

Merge Sort

List Size	Comparisons	Time (seconds)
1,000 (observed)	8703	0.002
2,000 (observed)	19409	0.003
4,000 (observed)	42774	0.005
8,000 (observed)	93591	0.01
16,000 (observed)	203267	0.02
32,000 (observed)	438603	0.045
100,000 (estimated)	1536618	0.102
500,000 (estimated)	8837121	0.459
1,000,000 (estimated)	18674232	0.988
10,000,000 (estimated)	220101127	13.940

1. Comparing the sorts at a general level, is one sort *always* better than the others?

   Merge sort is usually the better than the other insertion sorts and selection sorts on average compared to the other ones.

2. Which sort is better when sorting a list that is already sorted (or mostly sorted)?

Insertion sort has a best case of O(n) when most of the items are sort or all of them are sorted. While best case for selection is o(n^2) and merge sort is O(n logn).

3. You probably found that insertion sort has about half as many comparisons as selection sort. Why?

   There are certain cases where insertion sort doesn't have to go through every single object, while selection sort goes through every every single element, every single time. While Insertion sort operates differently

4. Given the above observation, why are the times for insertion sort not half what they are for selection sort? (For part of the answer, think about what insertion sort has to do more of compared to selection sort.)

   While insertion sort has half of the comparisons, there are still a lot of data moving, such as moving all of the elements over. Which takes a lot of time.